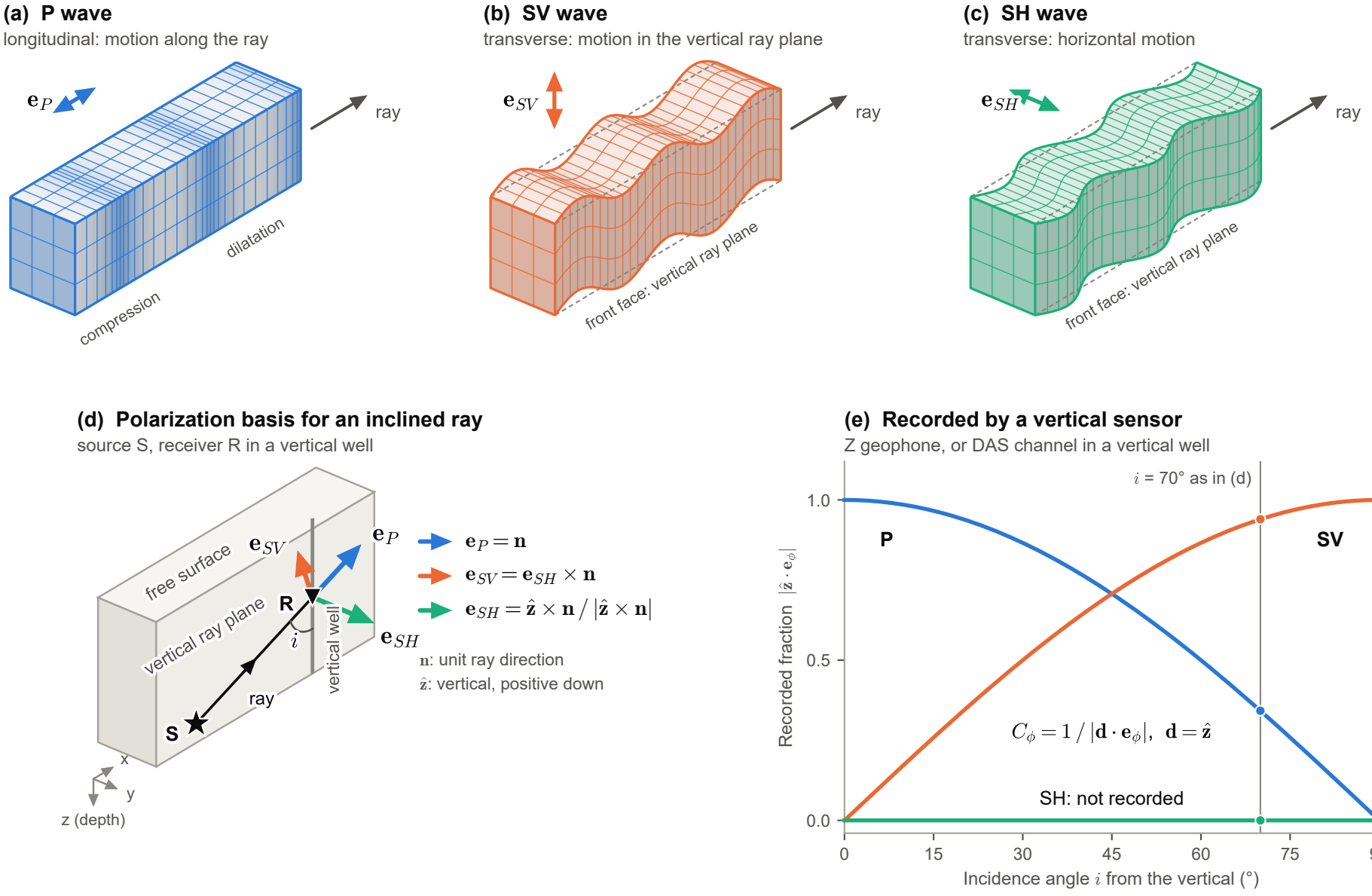

(a) P wave
longitudinal: motion along the ray
$\mathbf{e}_P$
ray
dilatation
compression
(b) SV wave
transverse: motion in the vertical ray plane
$\mathbf{e}_{SV}$
ray
front face: vertical ray plane
(c) SH wave
transverse: horizontal motion
$\mathbf{e}_{SH}$
ray
front face: vertical ray plane
(d) Polarization basis for an inclined ray
source S, receiver R in a vertical well
free surface
$\mathbf{e}_{SV}$
$\mathbf{e}_P$
vertical ray plane
R
i
vertical well
$\mathbf{e}_{SH}$
ray
S
x
y
z (depth)
$\mathbf{e}_P = \mathbf{n}$
$\mathbf{e}_{SV} = \mathbf{e}_{SH} \times \mathbf{n}$
$\mathbf{e}_{SH} = \hat{\mathbf{z}} \times \mathbf{n} \,/\, |\hat{\mathbf{z}} \times \mathbf{n}|$
$\mathbf{n}$: unit ray direction
$\hat{\mathbf{z}}$: vertical, positive down
(e) Recorded by a vertical sensor
Z geophone, or DAS channel in a vertical well
i = 70° as in (d)
1.0
0.5
0.0
P
SV
Recorded fraction $|\hat{\mathbf{z}} \cdot \mathbf{e}_\phi|$
$C_\phi = 1 \,/\, |\mathbf{d} \cdot \mathbf{e}_\phi|, \;\; \mathbf{d} = \hat{\mathbf{z}}$
SH: not recorded
0
15
30
45
60
75
90
Incidence angle i from the vertical (°)